

Master's Programme in Finance

Detecting Structural Evolution of Implied Volatility Surface Using Gradient-Based Features

Thesis subtitle (Machine Learning Approach to Market Regime Detection)

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Abstract

In this study, I aim to study how the implied volatility surface evolves over time by analyzing its structural changes using local gradients. I have developed a methodology to represent and quantify daily structural changes in the IV surface. I used a dataset of implied volatilities of S&P 500 index options across several moneyness and maturity levels and calculated changes in gradients at a set of points on the surface. I then used unsupervised clustering algorithm to identify distinct types of surface transformations based on these changes and movements in the index price.

My analysis revealed that there is a number of distinct types of surface transformations that can be identified and interpreted. Identified clusters represented specific skew or term structure dynamics for different levels of maturity and moneyness. My approach allows for flexible and model-free detection of IV surface transformations and allows to analyse the entire volatility surface.

Keywords Implied volatility surface, volatility regimes, options

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Preface and acknowledgements

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31 July 2025

Shakhin Dzhafarov

1 Introduction

In this thesis, I am studying structural evolution of the implied volatility surface. In order to present my idea in a clear and structured way, I will first describe several building blocks, including options, the Black-Scholes model, and implied volatility surface.

A call/put option is a right to buy/sell a specified amount of a financial instrument for a specific price at a specified point of time in the future (expiration date) for European options, or at any moment before the expiration date for American options.

In order to find the theoretical price of a European option, the Black-Scholes model (also known as the Black-Scholes-Merton model) is used. The following is the closed form solution for a European call option developed by Black and Scholes (1973):

$$C = S * N(d_1) - K * e^{-rT} * N(d_2) \quad (1)$$

$$d_1 = (\ln(S/K) + (r + (1/2) * \sigma^2) * T) / (\sigma * \sqrt{T})$$

$$d_2 = d_1 - \sigma * \sqrt{T}$$

where:

C = price of the European call option

S = current price of the underlying asset

K = strike price

T = time to expiration

r = risk-free interest rate

σ = volatility of the asset

$N(\cdot)$ = cumulative distribution function of the standard normal distribution

The Black-Scholes model has a few key assumptions, including constant volatility, constant risk-free rate, lognormally distributed price, absence of transaction costs and continuous trading. These assumptions are unrealistic, and a lot of research has been done to address them. For example, you could replace the Geometric Brownian motion with a jump-diffusion process to accurately model sudden jumps that happen in stock prices. You can also model the risk-free rate as a stochastic process.

The accuracy of these assumptions might vary depending on circumstances. For example, during news releases or corporate press releases prices can experience jumps, and the Geometric Brownian motion is far less adequate in this environment. Another example relates to the assumption of constant or deterministic interest rate. In case of short-term options, constant rate might be a realistic assumption, but in case of long-term options, you

can expect the rate to change over time, and the assumption becomes unrealistic.

My thesis most closely relates to the assumption of constant volatility. In practice, volatility is not constant and not deterministic. It can vary over time and react to market events and other factors. Therefore, we have to make a decision about how to model future volatility more accurately. We can classify existing approaches in the following way:

- **Historical volatility**
Under this approach, we calculate the historical realized volatility and assume that past volatility behaviour reflects future volatility behaviour.
This approach, however, has a couple of important limitations: it is backward-looking and does not reflect future market expectations. Moreover, it can be very slow to react to regime changes.
- **Implied volatility**
This is the volatility that's implied from an option price, using a pricing model, usually the Black-Scholes model. It is calculated numerically using, for example, Newton method or Bisection method. The idea is to tweak the volatility input in the model until we find the value of the volatility that makes the model price equal to the market price. The implied volatility that we determine in this way is forward-looking and theoretically reflects expectation of future volatility. We will discuss later, however, that this number can also reflect demand and other factors for a specific option that's not driven by future volatility expectations.
- **Model-based volatility**
This volatility is the output of a model that tries to capture how volatilities evolve using a stochastic or deterministic process. This approach requires making assumptions about the behaviour of volatilities over time and across strikes/maturities.

Historical and model-based approaches aim to forecast volatility using past data or structural assumptions. My thesis focuses on an empirical approach based on the implied volatility surface. This surface is a three-dimensional representation of implied volatilities across different maturities and moneyness levels. The figure below shows the implied volatility surface for S&p500 options on the 17 July 2025, the last day from the time period analyzed in this thesis (5 December 2007 – 17 July 2025):

IV Surface on 2025-07-17

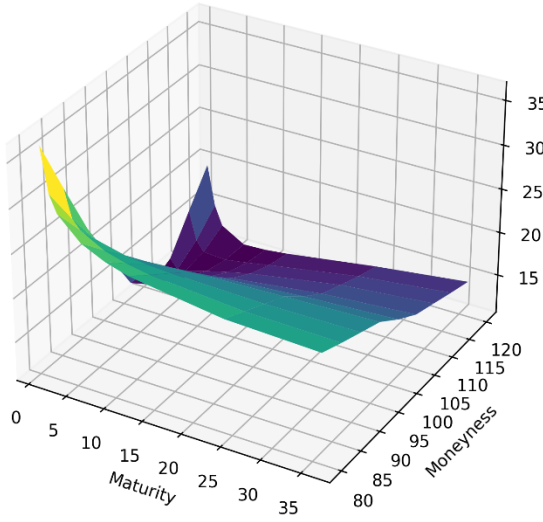


Figure 1. Implied volatility surface on 17 July 2025

The Figure shows the implied volatility plotted as a function of maturity (in months) and moneyness level. We can see that the volatility surface is far from being a flat plane, which contradicts the constant volatility assumption of the Black-Scholes model. This is a well-known phenomenon, and I will discuss it in more detail in the Literature Review section.

One of the details we can see in Figure 1 is that options of lower moneyness levels have higher implied volatilities than options of higher moneyness levels. We can also see that at-the-money options (moneyness 100) have the lowest implied volatilities, especially for options with shorter maturities. At the first glance, this could and should raise some questions. Since the volatilities in Figure 1 all represent expectation of future volatility for the same stock, shouldn't they all be equal, especially for a fixed maturity level? In theory, yes, but to understand why that is not the case, we have to take a look at how implied volatilities are calculated.

If we look at the Black-Scholes formula (1), (2), and (3), we can see that future volatility is the only variable we do not know and cannot observe in the market. We know the strike price, price of the underlying asset, time to expiration, risk-free rate, and we can observe the price of the option in the market. The idea is to find the volatility that makes the Black-Scholes model price equal to the observed market price.

We have to find that volatility numerically using, for example, Newton method or Bisection method since we cannot algebraically invert the Black-Scholes formula to solve for volatility. Newton method, for instance, is demonstrated below in Figure 2

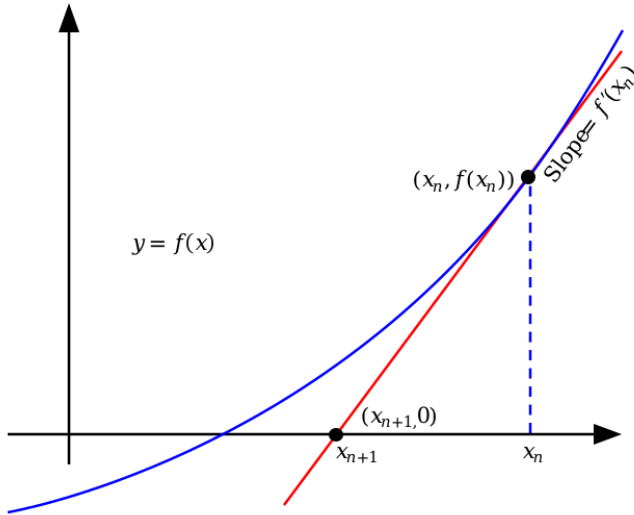


Figure 2. Newton's method (Wikipedia, 2025)

In the graph above, the x-axis represent volatility that we input into the Black-Scholes model and the y-axis represents the difference between the Black-Scholes model price and the market price. So, what we are trying to do is to find the volatility level that brings that difference sufficiently close to zero.

To be more precise, we take the following steps:

- Take an initial guess for the volatility, x_0
- Calculate the value of the function at x_0 and the slope (first derivative with respect to volatility) at the same point
- Find the next point x_1 using the following formula:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

x_1 will now be closer to the desired volatility value.

- Then we repeat the previous three steps with the new x until the difference between x_{n+1} and x_n is smaller than a predefined threshold.

We can see that, given a set of inputs such as strike price, price of the underlying, time to maturity and so on, the market price is what determines the implied volatility. This means that supply and demand for options of different strikes and maturities is what gives the implied volatility surface its shape.

It is useful to understand intuitively why higher expected volatility of a stock would make its options more valuable, and, inversely, why a high market price of an option would imply higher expected volatility. This comes

from the very nature of option contracts. As mentioned earlier, a European call/put option is the right to buy/sell a predefined amount of a financial instrument for a specific price at a specified point of time in the future. We only exercise a call/put option if the spot price at the maturity is higher/lower than the strike price, and in other cases the payout is 0. So, higher volatility positively affects the expected payout (payout weighted by probability of the spot being at a specific price at the maturity) because more extreme spot price outcomes (that are made possible by increased volatility) that are in-the-money are weighted by positive payout, whereas all the out-of-the-money outcomes are weighted by 0.

Moreover, increasing the volatility parameter of a lognormal distribution, which the stock price is assumed to follow under the Black-Scholes model, results in fatter tails of the distribution, which corresponds to higher probability of ending up significantly above or below the strike price. The weight of the right tail, however, expands more dramatically and the mean of the overall distribution shifts to the right, even though the peak value (mode) shifts to the left.

To visualize this, let's look at the following figure, which illustrates probability distribution functions (PDFs) for lognormal distributions with the same parameter μ but different parameters σ . These parameters are the mean and standard deviation of the underlying normal distribution, which is the distribution followed by the spot price under the assumptions of the Black-Scholes model:

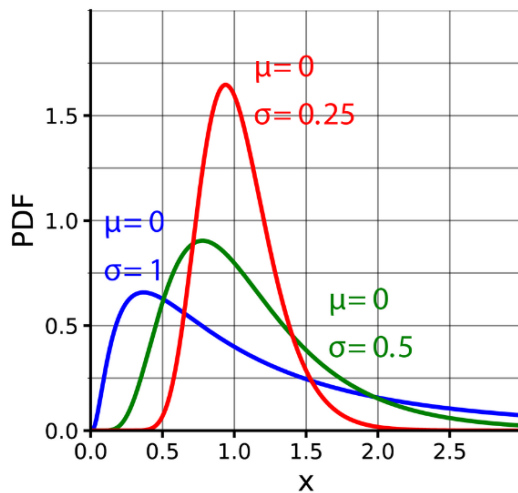


Figure 3. Log-normal distribution (Wikipedia, 2025)

We can see in the figure 3 that distributions with higher σ parameters have fatter tails. We can also verify that the mean of lognormal distributions increases when we increase σ increases. The expectation of lognormally distributed variable X is given by:

$$E[X] = e^{\mu + \frac{1}{2}\sigma^2}$$

We can see here that as we increase the value of σ , the value of $E[X]$ increases.

There are a few factors that affect the demand for options of specific moneyness and maturity levels and explain why the shape of the implied volatility is not flat and changes over time. Among those is the fact that options of different maturities cover different time periods, and market participants might expect different levels of volatility in different periods. This affects the term structure of volatility

Another reason is that, in contrast with the lognormality assumption of the Black-Scholes model, which implies normal distribution of returns, returns in practice are not normally distributed. Instead, their distribution has fatter tails which reflects a higher probability of extreme positive or negative returns than the normal distribution would imply. (Derman and Miller, 2006, p. 3) This increases the demand for options with strikes that are far away from the spot price. I will talk more about the factors affecting the shape of the volatility surface in the Literature Review section.

Below I will show a few volatility surfaces for dates when the shape of the surface had different characteristics. Let's consider a couple of dates from the 2008 financial crisis. The figure below demonstrates the implied volatility surfaces in the beginning of September 2008 and on 15 September 2008, which is the day Lehman Brothers filed for bankruptcy:

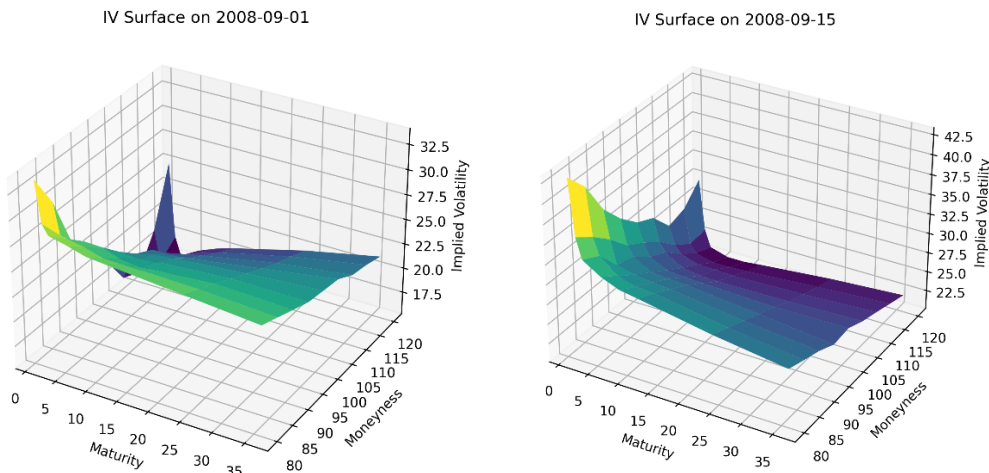


Figure 4. IV surfaces on 01 September 2008 and 15 September 2008.

We can see clear structural differences between the IV surfaces on 01 September and 15 September 2008. Most notably, implied volatilities for options with shorter maturities have increased drastically, which reflects the short-

term uncertainty triggered by the bankruptcy of Lehman Brothers. We can also observe that the area of the surface that saw the highest absolute increase in implied volatility is corner which represents lower moneyness levels and shorter maturities. This could be due to investors buying more protective put options in the time of crisis. We can also observe that options with longer maturities remained relatively stable and did not experience significant increases in implied volatilities. This could represent the market participants' belief that, at some point, the crisis would be resolved by government and/or central bank interventions.

Another example is the structural change in the volatility surface during the COVID-19 pandemic. The following figure shows IV surfaces in the beginning of March 2020 and on 16 March 2020:

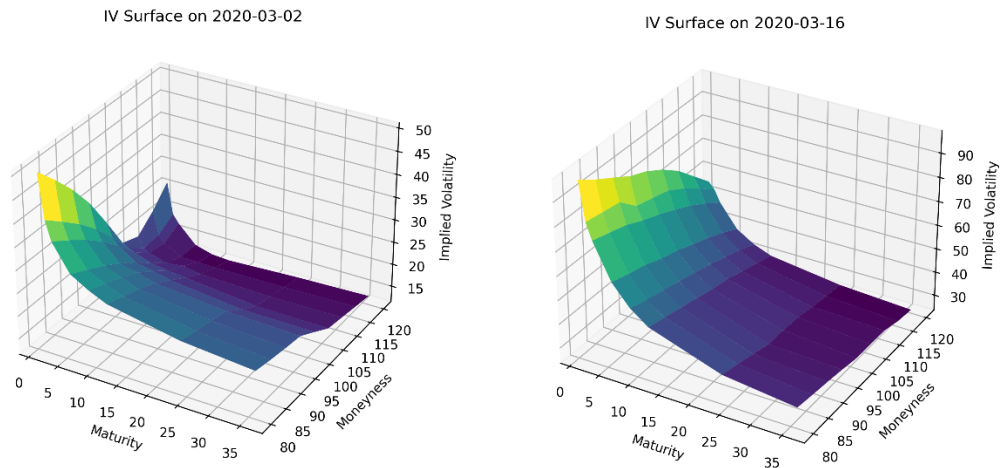


Figure 5. IV surfaces on 02 March 2020 and 16 March 2020.

We can notice a few key changes in the IV surface. We can see that volatilities across all moneyness and maturity levels increased dramatically, which reflects the uncertainty and panic in the stock market. We can also see that the typical volatility smile at short-term maturities disappeared and the volatilities of at-the-money options increased significantly relative to ITM and OTM options.

In the two above examples of IV surface transformations, we saw the structure of the surface change significantly, and we can see similarities and differences in these two cases. In both cases, we can get a good idea of how the market behaves and what it expects to happen by future maturity dates. In contrast, if we just looked at a general volatility level measure like VIX, we wouldn't get as much granularity and detailed view of the market's behaviour.

VIX is a popular measure of the market's expectation of volatility, and, like the IV surface we have discussed, it is calculated from S&P500 options. It is

often referred to as the fear index. (Wikipedia, 2025) The figure below shows the VIX index during the two time periods discussed above:

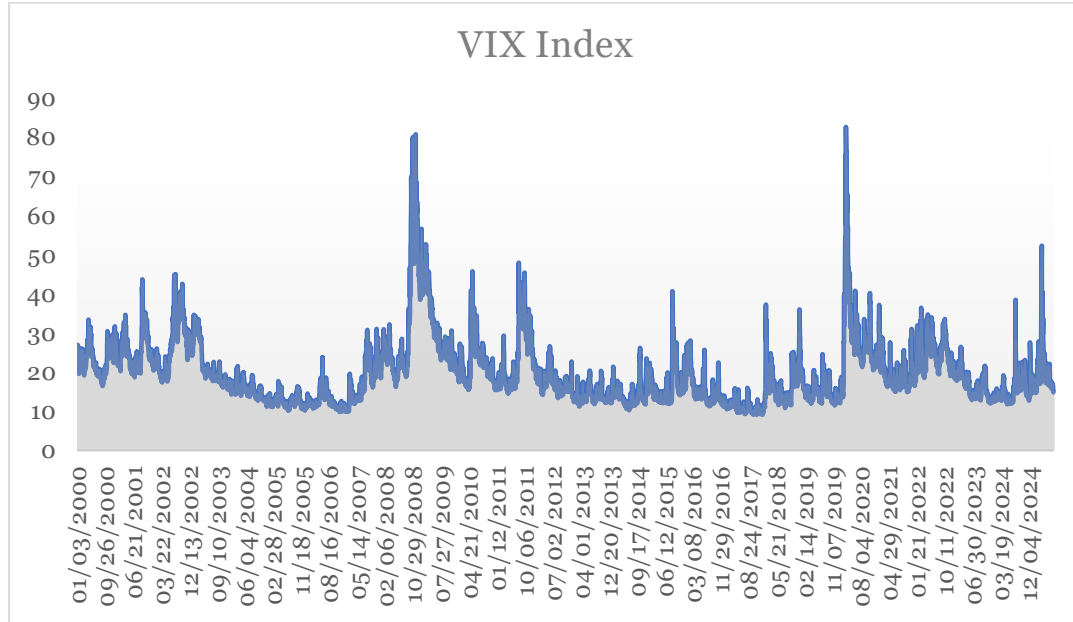


Figure 6. VIX Index, 2000-2025 (CBOE, 2024).

In the figure above, we can see spikes in volatility that correspond to the time periods we have discussed above. Often, VIX is used for defining volatility regimes and classifying periods into high and low volatility market regimes. These classifications are then used in risk management models, pricing models and other areas of quantitative finance where we need models to adapt to changing market conditions.

The applications where regimes play an important role include:

- Regime-switching models, where the behaviour of the model depends on the current regime. Examples include Markov regime-switching models.
- Filtering historical data to check statistics under certain market conditions, or to train models separately for high and low volatility environments.
- Adjusting parameters in risk management and hedging models based on the current regime. This way, the models can be more realistic and accurate.
- Backtesting trading strategies, for which the performance during specific kinds of market conditions is important. Moreover, certain strategies might only be designed to trade in certain regimes.
- Portfolio allocation, when we want to adjust our exposure to different asset classes based on the current market regime.

- Stress testing. We can define types of regimes and simulate performance under each of them.

When we define regimes based on the overall level of volatility, which is calculated from options with a limited set of moneyness and maturity levels, we get a good, interpretable summary number that is easy to use and represents overall market sentiment. However, we miss important information that is contained in the volatility surface. For example, looking at volatility skew, smile, and the term structure of volatility gives us a good idea of investor behaviour, risk perception, and market expectations. Even looking at anomalies in implied volatilities of options of specific moneyness and maturity levels can, in some cases, be informative if combined with other data and knowledge explaining why could such an anomaly occur.

In my thesis, I would like to define regimes based on the structure of the entire volatility surface. As I have mentioned, a major benefit of doing so is that we get more information on different IVs, which lets us define regimes not just by how high or low volatility is, but how volatilities behave across different maturities and moneyness levels.

One example where it would be useful to use this additional information is a situation where two periods have the same VIX level, but in one case the volatility skew is flat, and in the other case it's downward sloping. These two cases represent different investor behaviour, but a regime definition based on the VIX level would classify them into one regime group.

Defining regimes based on the IV surface can also improve performance of models that try to capture the behaviour of path-dependent and exotic options. Moreover, it can improve development, testing, and implementation of trading strategies and risk management models.

In my thesis, I am interested in how the surface evolves over time. I take an empirical approach to find if there are recurring types of structural changes in the IV surface. I will describe my methodology in more detail in Section ... Essentially, I quantify a transformation at a specific moneyness-maturity point of the surface as the daily change in gradient at that point. I do the same for a set of points on the surface. These data points together tell us how the structure of the IV surface changed from one day to another.

Changes in the IV surface are intertwined with movements of the stock price, so I also include it in my analysis. Traditional approach to modelling how the surface moves in response to movements in the price of the underlying is to make an assumption such as sticky strike or sticky delta. These approaches differ in terms of what volatilities stay constant as the underlying moves.

Under the sticky strike approach, it is assumed implied volatility at a given strike price remains constant as the underlying moves. Sticky delta, on the other hand, assumes that IVs at a given moneyness level remain constant as the underlying moves. In my thesis, however, I do not make such

assumptions as they might not hold across time and under different market conditions. Moreover, these approaches imply that the behaviour of the surface is the same regardless of where we are on the surface. However, this might not be the case and volatilities at different points of the surface might behave differently. In my thesis, I take an empirical approach by measuring the actual changes in the surface and the spot price and looking for patterns and distinct types of surface transformations.

My goal in this thesis is to study how the implied volatility surface evolves over time by analyzing its local structural changes. More specifically, I aim to:

- Develop a methodology to represent daily structural changes in the IV surface using gradients at a set of points on the surface.
- Identify recurring surface transformation patterns using unsupervised machine learning.
- Interpret and classify the identified surface transformation types.

Primary research question:

“How can structural changes in the implied volatility surface be quantified and classified into distinct groups using gradient-based features and machine learning techniques, and what do these changes tell us about market dynamics?”

Subordinate questions:

- How can a volatility surface transformation be quantified?
- Can clustering methods reveal distinct types of structural surface changes?
- Do identified clusters align with known market regimes and what is the economic explanation for them?

Scope and limits of the study:

- I use data on S&P 500 options. I use historical data across different moneyness and maturity levels ranging from 2007 to 2025.
- The data is of daily frequency, so the calculated changes in gradients are also daily. Therefore, I focus on studying short-term changes in the surface.
- Regimes are detected from data empirically, and not using a set of pre-defined criteria or models.
- As the IV data is for a set of discrete maturity and moneyness levels, I use a finite-difference method to approximate local gradients.

Next is the Literature Review section, where I present and summarize relevant research on implied volatility surfaces, volatility regimes and regime-

switching models. In Section 3 I describe the data and methodology used in more detail. In Section 4 I present the results of my study, after which I summarize and conclude my thesis in Section 5.

2 Literature review

In this section, I will present existing research on implied volatility surfaces, volatility regimes and regime-switching models. I will start with ...

2.1 Implied volatility and IV surface

As I have mentioned in the Introduction section, implied volatilities are calculated numerically from the Black-Scholes formula, which was introduced by Black and Scholes in 1973. Black and Scholes (1973) derived a closed-form formula for the pricing options using the no-arbitrage principle:

$$C = S * N(d_1) - K * e^{-rT} * N(d_2) \quad (1)$$

$$d_1 = (\ln(S/K) + (r + (1/2) * \sigma^2) * T) / (\sigma * \text{sqrt}(T))$$

$$d_2 = d_1 - \sigma * \text{sqrt}(T)$$

This formula provides a closed-form expression for the option price as a function of several variables, including volatility, so it can be used to numerically find implied volatility for a given option, which lets us construct the implied volatility surface. Specifically, we find IVs by setting the model price equal to the market price and numerically finding the volatility that makes the two sides of the equation sufficiently close to each other. We can use, for example, Newton's method or Bisection method for this purpose. While the surface should be flat under the assumptions of the Black-Scholes model, it is not flat in practice.

Derman and Miller (2016) note that we volatility smile is an observable feature of the implied volatility surface ever since the 1987 stock market crash. Volatility smile is a pattern in the implied volatilities for a given maturity, where we see a smile shape in IVs when we plot them against strikes or moneyness. The figure below illustrates how implied volatilities looked like as function of moneyness (strike/index price) for S&P 500 options prior to 1987:

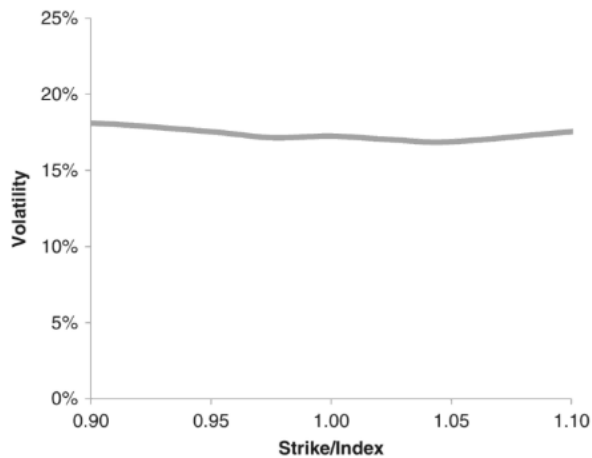


Figure 7. Typical implied volatility pattern prior to 1987 stock market crash (Derman and Miller, 2006, p. 4).

We can see that the IV curve is almost flat as a function of moneyness, which is different from the volatility smile that has been the typical shape ever since 1987. Derman and Miller (2006, p. 4) note that the shape is actually more of a smirk, which means that the smile has a negative skew. The term structure of volatility is also a deviation from the assumptions of the Black-Scholes model as volatility is assumed to be constant over time (Derman and Miller, 2006, p. 144).

2.2 IV surface movements

The IV surface moves in response to changes in demand patterns across maturities and moneyness levels, as a result of macroeconomic or other significant events, or in response to movements in the price of the underlying. Derman (1999) studies how the IV surface moves as a result of spot movements and notes that there are many suggested rules and theoretical models that attempt to describe how the IV surface moves and how the skew varies, but they lead to different predictions, and a lot of them do not perform well. One of the patterns that researchers have identified is that implied volatilities are negatively correlated with movements of the index price. Derman (1999, p. 3) illustrates this relationship with the following graph:

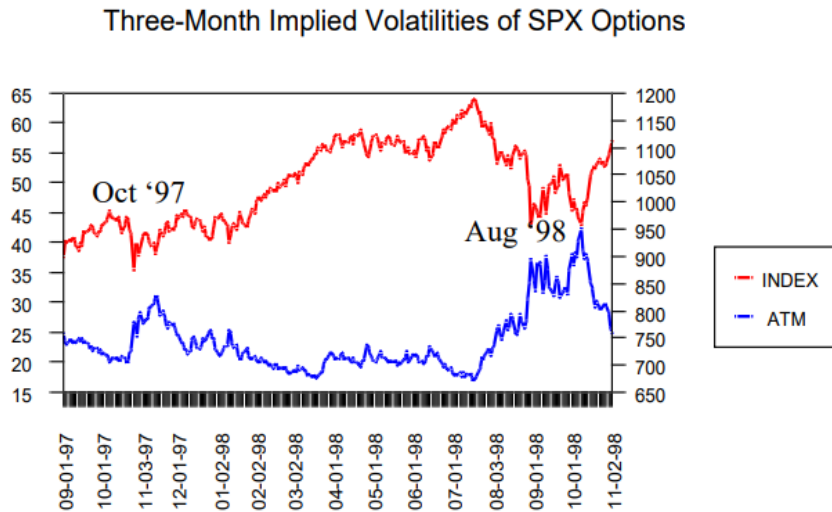


Figure 8. S&P 500 index price and its three-month IVs (Derman, 1999, p.3).

We can clearly see the negative relationship between the index price and implied volatility. It is important to note, however, that this relationship only seems to be strong for indices, and not for individual stocks.

Hull and White (2016, p. 181) describe the two main hypotheses that aim to explain this phenomenon. One hypothesis, which is called the leverage effect, is based on the idea that when prices move down, leverage increases, which leads to an increase in volatility. In the same way, when prices move up, leverage decreases, which results in a decrease in volatility. (Hull and White, 2016, p. 181) Christie (1982, p. 427-428) attributes this relationship to leverage effect and shows that volatility is an increasing function of leverage.

The other hypothesis states that when volatility level changes, the required rate of return changes accordingly. So, when the volatility increases, investors require a higher rate of return, which leads to a decrease in price, and vice versa. (Hull and White, 2016, p. 181) This phenomenon has been studied by researchers, and, according to Hull and White (2016, p. 181), the evidence seems to support the volatility feedback effect more.

For example, French et al. (1987, p. 27) find evidence that when volatility is predictably high, investors demand higher returns to compensate for taking the extra risk. Moreover, they find a strong negative correlation between unexpected increases in volatility and stock returns. Campbell and Hentschel (1992) use the QGARCH model, which allows to capture the negative correlation between movements in price and volatility, on U.S. stock return data and find that incorporating this empirical fact into the model improves it significantly. These findings support the volatility feedback effect hypothesis.

Derman (1999, p. 4) studies the behaviour of implied volatilities during the period from 01 September 1997 to 02 November 1998 for options of fixed

strikes ranging from 750 to 1200, and divides the period into 7 different regimes. The regimes are determined and classified qualitatively based on observed patterns. They are shown below in Figure 9:

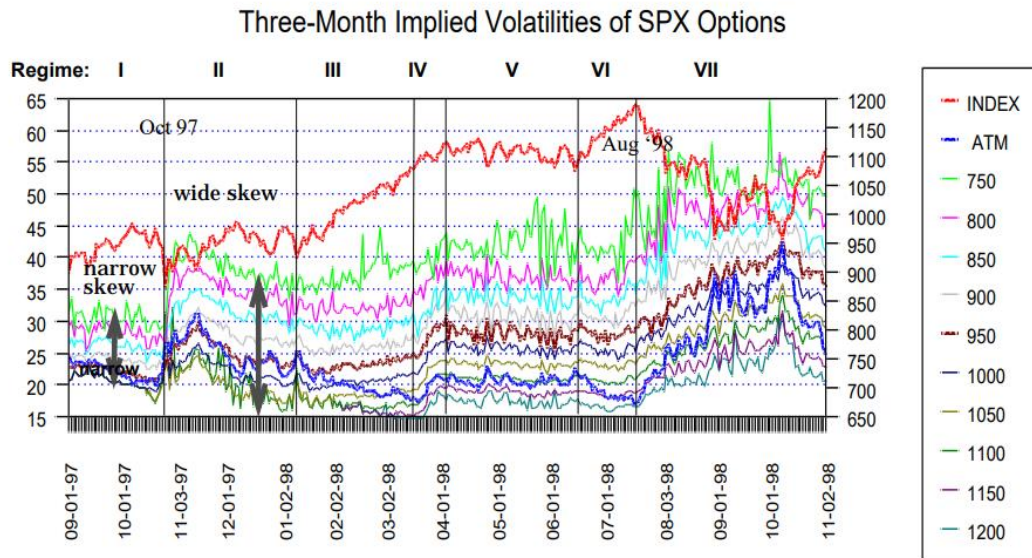


Figure 9. S&P 500 index price and its three-month implied volatilities for ATM options and for options with strike prices ranging from 750 to 1200 (Derman, 1999, p. 5).

We can see in the figure above that IVs increase as we look at options with lower strike prices. This means that the volatility skew is negative during the entire period. We can also judge how steep the slope of the skew is by looking at how spread out the graphs of volatilities are for low and high strike prices. The figure above also includes arrows, which represent a measure of skew magnitude. (Derman, 1999, p. 5) We can see that the magnitude of the skew has increased significantly over time.

The figure above also includes the identified regimes. They have distinct features and patterns of behaviour of the implied volatilities. We can see the following regimes:

- Regime 1: index price increasing, implied volatilities had a slight downward movement
- Regime 2: index price fell sharply, realized and implied volatilities increased, the volatility skew widened as we can see from the arrow on the chart
- Regime 3: stable increase in the price of the index, implied volatilities at fixed strike levels remained stable. Note that the ATM implied

volatility gradually decreased in this period because as the index price increased, the strike price of ATM options increased as well

- Regime 4: Derman contrasts this regime to regime 3 because during both regimes the index price had a slow and stable increase. However, unlike in the third regime, IVs increased by a few volatility points
- Regime 5: stagnant index price and implied volatilities
- Regime 6: both the index price and volatilities behaved similarly to regime 3
- Regime 7: index price fell significantly and had a recovery towards the end of the period, realized and implied volatilities increased. Derman compares this regime to regime 2 because in both periods IVs moved inversely with the index price. (Derman, 1999, pp. 4-6)

These regimes have the following annualized realized volatilities:

Regime	Realized volatility
1	17
2	21
3	12
4	9
5	13
6	11
7	30

Table 1. Realized volatility by regime (Derman, 1999, p. 6)

We can notice that IVs had quite distinct behaviours under some regimes. Therefore, classifying them into distinct groups can be beneficial for various applications. In the paper, Derman explores how the IV surface moved in response to spot price movements. Specifically, he explores whether “sticky strike”, “sticky delta”, or “sticky implied tree” rule works in each regime. These rules determine how implied volatilities move when the spot price moves.

These rules differ in terms of what stays constant, or “sticky”, as the spot price moves. “Sticky delta” means that IVs for options of fixed delta levels stay constant. On the other hand, “sticky strike” means that as the spot price moves, IVs at fixed strike prices remain unchanged. In the “sticky implied tree” model, we construct a tree of local volatilities that is based on current options of all strikes and maturities. This implied tree stays constant as the spot price moves. To illustrate this, Derman (1999, p. 13) shows the following graph:

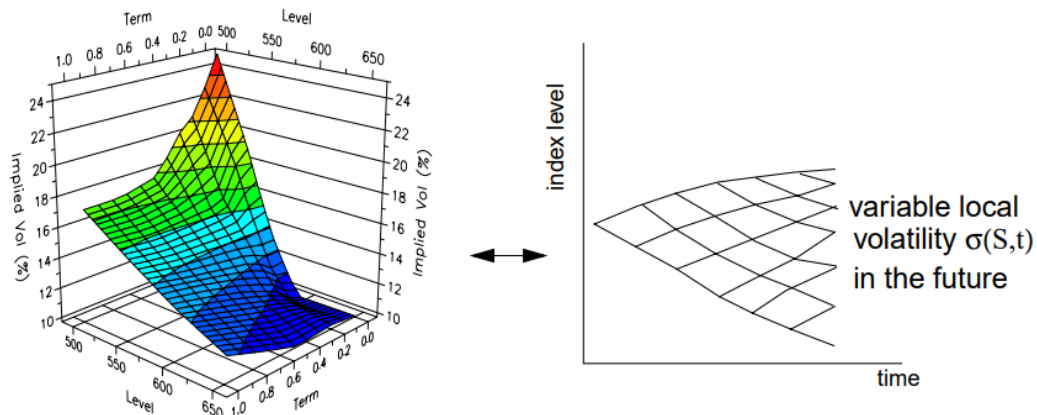


Figure 10. Volatility surface and its implied tree (Derman, 1999, p. 13).

We can see in the figure above that the surface is transformed into a tree, where the dependence on the time variable is implied from options of different maturities, and the dependence on the spot price is implied from options of different strikes.

Correctly classifying each regime and determining which rule works best during each one is important for:

- accurate options pricing models and repricing options after spot moves
- risk management and hedging, for example, when the hedge ratio is influenced by volatility
- stress testing for different scenarios

Derman (1999, p. 23) finds that during stable regimes 1 and 5 and stable growth regimes 3 and 6, sticky strike rule applied, whereas during regimes 2 and 7, where we saw increased volatility of returns and downward movements, the implied tree model applied.

My thesis is also based on the idea that the behaviour of the implied volatility surface changes over time, but there are some recurring patterns and certain types of transformations that we can detect and explain. Another similarity is that Derman also looks at changes in volatilities rather than looking at snapshots of the volatility skew. At the same time, my thesis is different in that I do not make any assumptions about stickiness rules. Instead, I take an empirical approach. Moreover, I take a machine learning approach to detecting patterns in daily surface transformations.

There are other studies that covered regimes and regime switching models. For example, Hamilton (1989) introduces a Markov-switching autoregressive model that switched between discrete regimes, which allows the model to adjust itself when structural changes occur in the market.

My thesis is also based on the idea that economic and financial time series data can switch between different behaviours, and that models should be flexible enough to capture this. At the same time, my thesis is different from Hamilton's study in that I make no regime assumptions, instead relying on day-to-day observed changes in gradients at key points on the surface and unsupervised learning for detecting distinct types of regime transformations. Moreover, my approach is to study changes in the structure and classify transformations, or movements of the surface, rather than regimes themselves. Another distinction is that Hamilton analyses macroeconomic data, whereas I analyse implied volatility data.

There is also an article by Maven Securities titled "Implied Volatility Surface Regimes" which touches on some of the ideas that I'm exploring on my thesis. In the article, the author talks about identifying regimes using clustering based on ATM volatility, skew, and smile. Below is the illustration of STOXX50E implied volatility features provided in the article:



Figure 11. ATM Volatility, Skew, and Smile of STOXX50E implied volatility (Maven Securities, 2021)

It is noted in the article that we do not see apparent shifts between states on this plot. However, looking at the data from a different perspective and clustering the data based on three dimensions mentioned above (ATM vol, skew, and smile) helps the author identify three distinct regimes. The figure below illustrates the time series of identified regimes:



Figure 12. Regimes plotted over time (Maven Securities, 2021)

The figure above illustrates the identified regimes and makes it easy to see transitions between regimes. The article also provides scatter plots illustrating distinct features of the three clusters and links them to macroeconomic events and trends.

In my thesis, I also perform feature-based regime detection using implied volatility features and clustering. My approach, however, differs in a few ways. First, I am using data across the entire volatility surface, including different moneyness and maturity levels, looking for patterns in different areas of the surface. Moreover, the feature vector that I am feeding into an ML model is higher dimensional as I consider many points across the surface, although I do use principal component analysis to reduce dimensionality. I am also using a different clustering method. What is more, I am focusing on day-to-day changes in the surface to identify structural shifts in the IV surface, which might correspond to shifts between regimes, rather than clustering regimes themselves.

To summarize, the importance of this area of research comes from the fact that, contrary to the Black-Scholes model assumptions, volatility is not constant, and the implied volatility surface is not flat. This makes it necessary for us to understand and model how the surface evolves and what are the reasons behind it. In this literature review, I have covered different stickiness rules that aim to capture how IVs move when the spot price moves. I have also talked about the negative correlation between movements in the index price and volatility. An important part of this literature review was the overview of relevant research on regimes, their characteristics, as well as how they can be applied in regime-switching models.

The approach that I take in my thesis has some key differences from the existing research. First, I take into account the entire volatility surface and quantify structural changes using local gradients. Second, I do not make any assumptions neither about how the IV surface behaves nor about regimes

that I expect to see in the data. Instead, I take an empirical and machine learning approach to detect and classify recurring patterns of surface transformations. Another key difference is that I focus on daily changes in the IV surface rather than daily snapshots of the IV surface, which means that I am looking to find recurring patterns of surface transformations, which may or may not correspond to transitions between regimes, rather than looking for regimes themselves.

3 Research material and methods

In this section I will talk about the methodology I used in this study. I will describe the data I used, as well as my approach in quantifying the structural changes in the implied volatility surface. I will also describe the machine learning approach that I took and the feature vector that I've constructed to be fed into the ML model.

3.1 Data

The data I used in my thesis comes from Refinitiv. The first step was to retrieve the data on S&P 500 index options. I chose to construct the implied volatility surface based on moneyness and maturity levels. Therefore, I retrieved Refinitiv's data on the index options by moneyness level, ranging from 80% to 120% moneyness. For each moneyness level and for each date, I used implied volatility data for 8 maturity levels (1, 2, 3, 6, 9, 12, 24, and 36 months).

The data is in the form of synthetic continuous (non-expiring) contracts that are constructed from actual contracts. This makes it possible to perform time-series analysis. Using data with consistent maturity and moneyness levels ensures that we can compare and analyse the data across time. The dates range from 5 December 2007 to 17 July 2025. In addition to the IV data, I also retrieved data for S&P 500 spot price for the same period.

Once I retrieved the data through MS Excel Refinitiv add-in, I converted it into csv files and imported to Python. Inside of Python, I mainly used the pandas library to handle the data.

3.2 Implied volatility surface

I stored all the data in a dataframe. When writing code, I needed to change the format from one to another or tweak the format of the original dataframe for specific tasks. For example, sometimes it was more comfortable to use the long format for a dataframe, where each row corresponds to a specific date, moneyness, and maturity, and sometimes it was better to represent the data in a wide format, where each date has its own row and the columns represent a point on the volatility surface (a specific level of moneyness and maturity).

Remember that the data I retrieved was for a set of specific maturity and moneyness levels. A common thing to do is to interpolate the entire surface from that set of points if you need to know the volatility for an arbitrary point on the surface that's not part of the dataset, using, for example, cubic spline interpolation. However, I did not need to do it in my thesis because my approach is based on gradients at a discrete set of points, or a discrete

moneyness-maturity grid. This also helps to keep my approach empirical without making any interpolation assumptions.

3.3 Gradients

As I've mentioned, my approach was to calculate gradients on a set of points on the IV surface and compute daily changes in those gradients. It is useful to think about the gradient at a point as a vector that's perpendicular to tangent plane to the IV surface at that point. The figure below illustrates this idea:

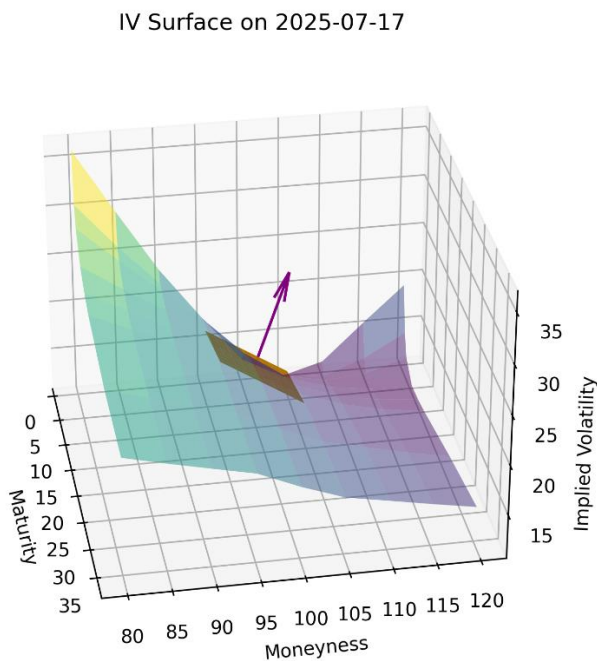


Figure 13. IV Surface on 17 July 2025.

In the figure above, I have visualized the tangent plane to the volatility surface at the 100% moneyness level and 9-month maturity level. I have also included the gradient at that point and scaled it to make it better visible on the plot. The idea is that when the local behaviour of the volatility surface changes, the direction and magnitude of the gradient will change, and finding the difference of gradients at different points on the surface represents the overall structural change.

The figure below illustrates how the tangent plane and gradient direction changed in 3 months from 17 April 2025 to 17 July 2025:

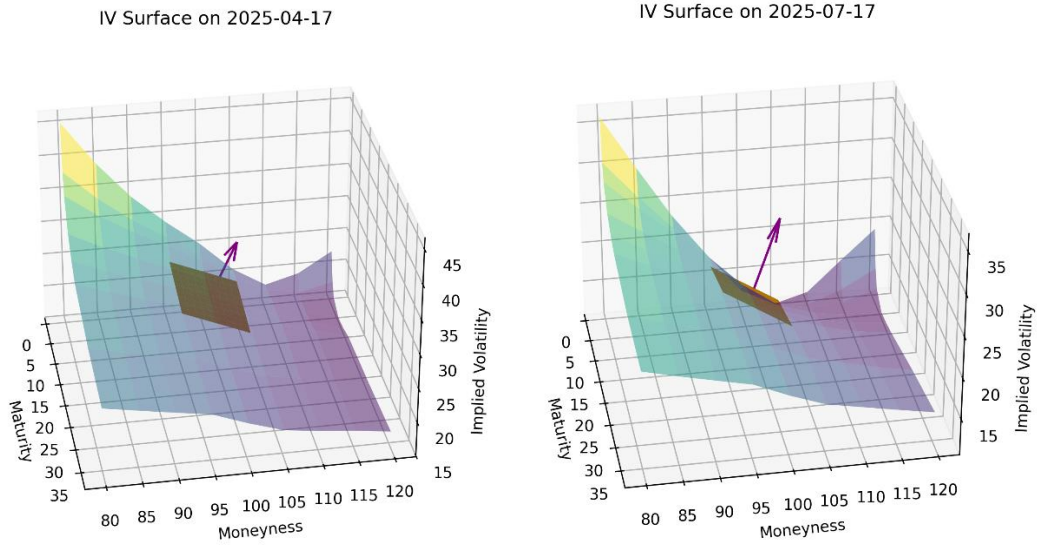


Figure 14. IV surfaces on 17 April 2025 and 17 July 2025.

We can see here that the orientation of the tangent plane has changed considerably. I have provided this example with two dates that are 3 months apart from each other to make the change in the direction of the gradient vector more visible. However, it is important to note that I study daily changes in gradients.

With this approach, I do not make any assumptions about how the IV surface moves, instead capturing the actual movements from historical data. There are overall 72 points on the surface (9 moneyness levels times 8 maturity levels) for which we compute daily gradient changes. Each gradient consists of two components, one is the partial derivative of implied volatility with respect to moneyness, and the other is the partial derivative with respect to maturity. Therefore, there are 144 gradient-related features for each day. Later I will talk about using principal component analysis to reduce the number of features.

3.4 Spot price feature

One more feature that I added to the feature vector is the daily relative change in the spot price of the S&P 500 index. As I have mentioned earlier, there is in practice correlation between movements in the index price and volatility, and adding this feature could capture this. Moreover, clustering could detect, for example, sharp increases in volatility, skew shift, or other structural changes that are accompanied by movements in the spot price. Moreover, it could detect differences between surface transformations during days with positive and negative returns.

3.5 Principal component analysis

Before conducting further analysis, I convert the data into a wide format, where each row corresponds to a date and each column corresponds to a point on the volatility surface with a given moneyness and maturity. Actually, there are two columns per points, one representing the change in the moneyness component of the gradient, and one representing the maturity component of the gradient. I also scale the data so that the features are on a comparable scale.

Next, I use principal component analysis (PCA) to reduce dimensionality. PCA reduces the number of correlated variables into a smaller set of uncorrelated, or less correlated variables. Our goal is to explain a high percentage of variance while reducing the number of features to be fed into the ML model. For the PCA, we have to choose the number of principal components to reduce our variables into. The figure below illustrates how much of the variance in the data we can explain as a function of the number of principal components we choose:

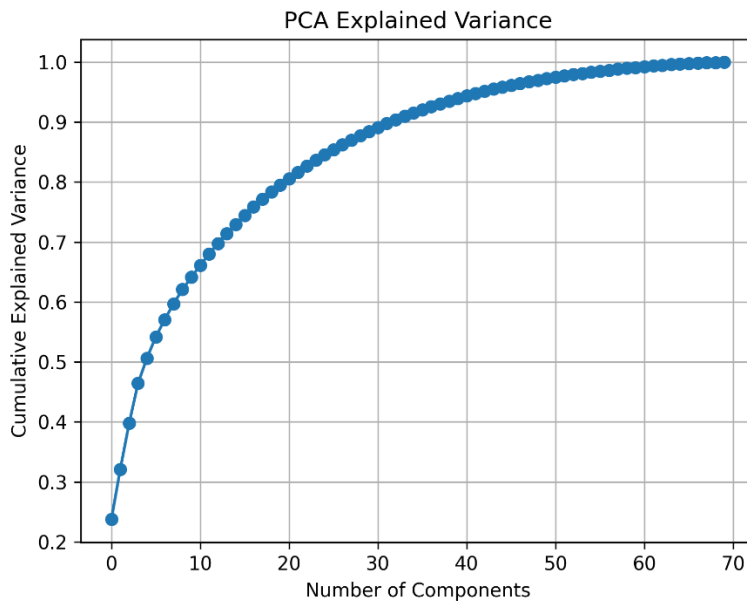


Figure 15. PCA explained variance as a function of the number of components.

We can see in the figure above that 30 principal components explain nearly 90% of the variance, and 40 components explain nearly 95%. I have tried setting the number of components to different values, including 30, 40, and 50, and all options lead to quite similar clusters when applying the ML model.

3.6 Clustering surface transformations

The next step is to use a clustering algorithm to identify types of surface transformations that have similar characteristics. I use Hierarchical Density-Based Spatial Clustering of Applications with Noise, or HDBSCAN, which is an unsupervised learning algorithm for clustering data. I chose it for several reasons:

- It does require specifying the number of clusters in advance, which is useful in this case because I don't want to assume the number and types of distinct structural transformations of the IV surface
- The detected clusters are stable and persist even as I change some parameters or tweak the input feature vector
- Explicitly labels noise, which is useful in this case because, intuitively, we can expect changes in the IV surface during most days to look like random noise
- It captures clusters of arbitrary shape

An important parameter was the minimum cluster size, which determines how large a cluster has to be. Decreasing this parameter led to identifying more clusters, but what's important is that the clusters that were found using a higher values of this parameter remained unchanged when decreasing it.

The next step was to visualize and interpret the clusters, which I will talk about the next section.

4 Results

As expected, most of the daily surface transformations were classified as noise. However, I found several clusters with distinct and recurring types of structural changes. There were 13 distinct clusters with varying sizes, ranging from 5 to 18 samples each. In this section I will describe three clusters, and I will include the heat maps for other clusters in the appendix. For each cluster, I will show two heat maps, one for the moneyness component of the gradient, and one for the maturity component, for each point on the volatility surface.

4.1 Cluster 1

The figure below illustrates the heat map of the gradient components across moneyness and maturity levels:

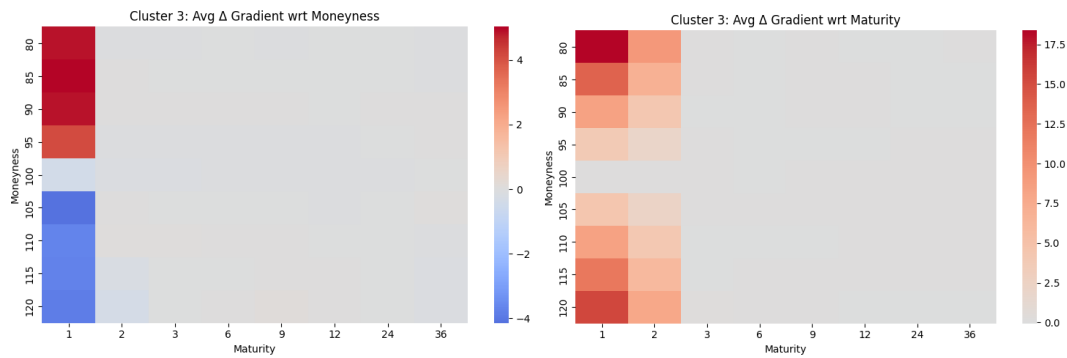


Figure 16. Gradient heat map for cluster 1

From the heatmaps above, we can see that this transformation corresponds mainly to short-term implied volatilities, for options with maturities of 1 and 2 months.

We can also see that the change in the moneyness component of the gradient is strongly negative for high moneyness and strongly positive for low moneyness. This corresponds to a flattening volatility skew. In other words, the volatility smile becomes less pronounced. From the maturity component's heatmap, we can see that gradients increase, which means that short-term portion of the volatility surface falls while the long-term portion stays relatively stable.

This suggests that this structural change can be interpreted as an expectation of stability in short-term and less bearish expectations.

4.2 Cluster 2

Below we can see the heat maps representing the second cluster:

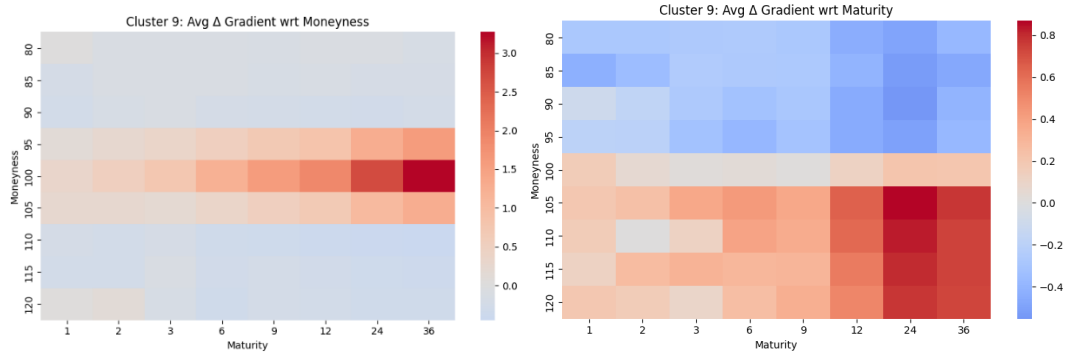


Figure 17. Gradient heat map for cluster 2

Here we can see from the maturity component's heatmap that for low moneyness options, the term structure of IVs has transformed in the following way. For all maturity levels, the maturity component of the gradient decreased, which means that IVs of longer-term maturities have fallen relative to shorter term maturities. For example, if we assume that the term structure had a negative slope prior to this transformation, then after this transformation the slope would be more negative, or in other words, the term structure would be steeper.

On the other hand, looking at the same heatmap on the right-hand side, we see a different picture for high moneyness, where term structure shifter the other way around, which means that longer-term volatilities decreased relative to shorter-term volatilities.

The behaviour on the two opposite side of the moneyness scale suggests negative change in the short-term expectations that is accompanied by positive long-term expectations. Since volatilities for low moneyness are driven by OTM puts and volatilities for high moneyness are driven by OTM calls, this structural shift seems to suggest that investors are pricing in short-term downside risk. We can explain the shift for high moneyness options in the following way: since there is short-term bearish outlook, the demand for short-term OTM calls (which drive volatilities for high moneyness levels) decreases, while the demand for longer-term OTM calls stays constant or increases (since this transformation reflects rise in short-term uncertainty or short-term bearish outlook).

4.3 Cluster 3

The figure below shows the heat map for the third cluster:

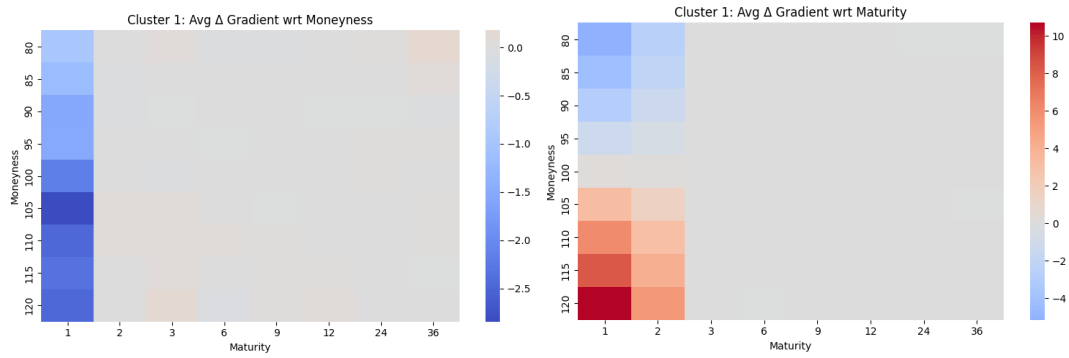


Figure 18. Gradient heat map for cluster 3

In this cluster, we can see from the heat map on the left side that for all moneyness levels of short-term options, the moneyness component of the gradient decreased, which means that higher moneyness volatilities have fallen relative to lower moneyness volatilities. This corresponds to a decrease in the slope of the skew. This seems to suggest that demand for OTM puts increases relative to demand for OTM calls, which suggests that investors are becoming more bearish.

5 Summary

My goal was to analyze structural changes in the implied volatility surface and study how the surface evolves over time. In this study, I have successfully developed a methodology to represent and quantify daily structural changes in the IV surface using local gradients. I have also used unsupervised clustering algorithm to identify distinct types of surface transformations. I have also included interpretations for some of the clusters in the previous section.

My analysis revealed that there is a number of distinct types of surface transformations that can be identified and interpreted. As expected, the vast majority of daily IV surface changes were classified as noise because structural changes in the volatility surface do not occur often. Therefore, the identified clusters have sizes of up to 18 samples. Notably, several clusters represented specific skew or term structure dynamics for different levels of maturity and moneyness.

My approach allows for flexible and model-free detection of IV surface transformations and allows to analyse the entire volatility surface. There are, however, limitations. For example, I have focused on daily surface transformations, and further research could study longer-term transformations. Another limitation is that the interpretation of the identified clusters may be subjective.

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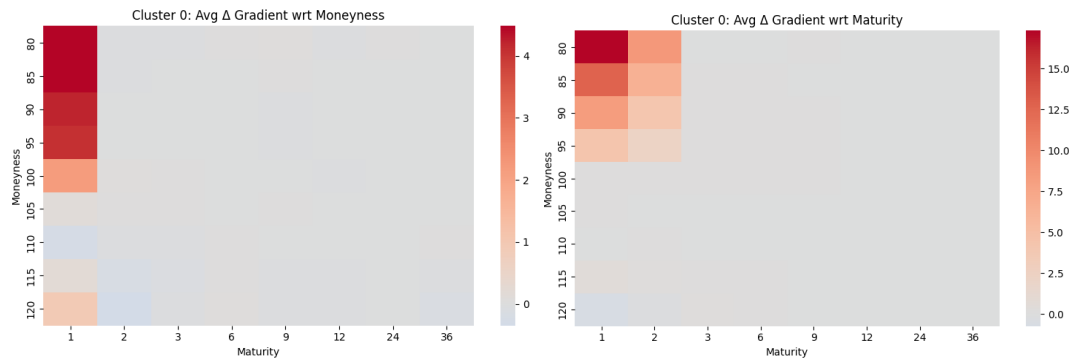
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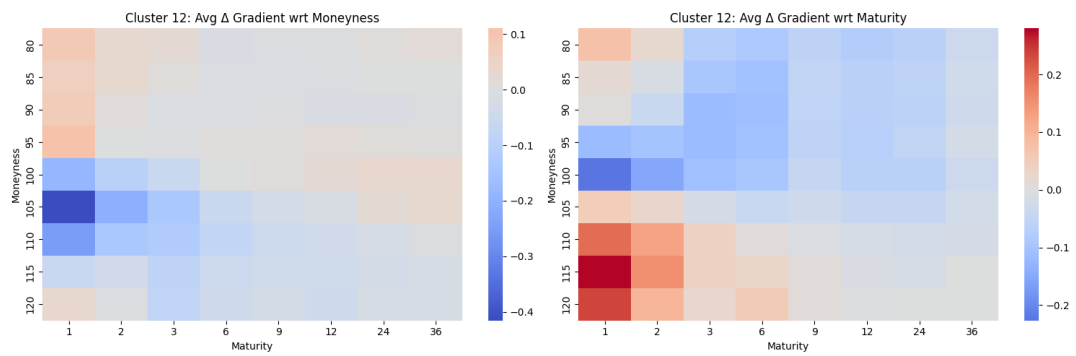
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A. Clusters

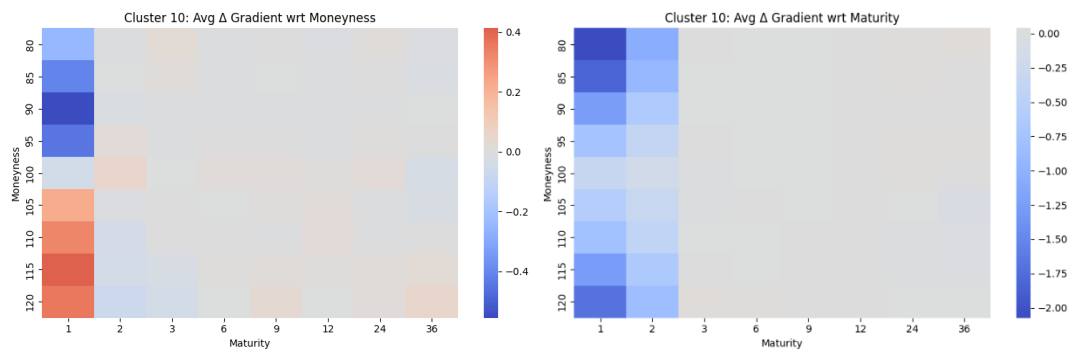
Cluster 4



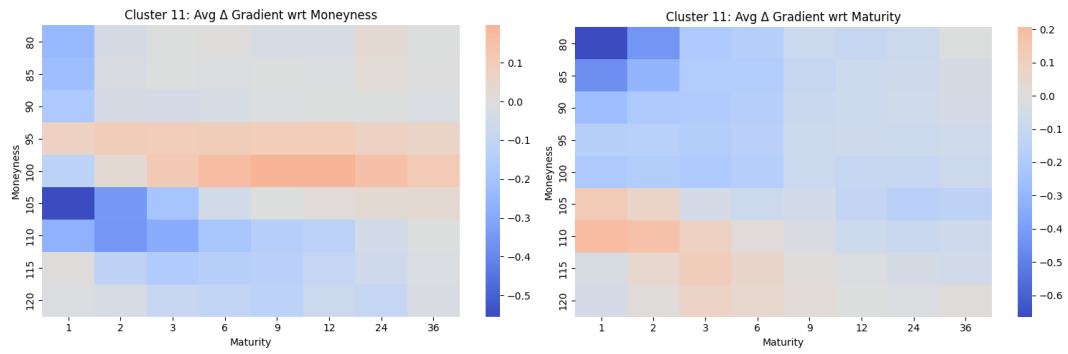
Cluster 5



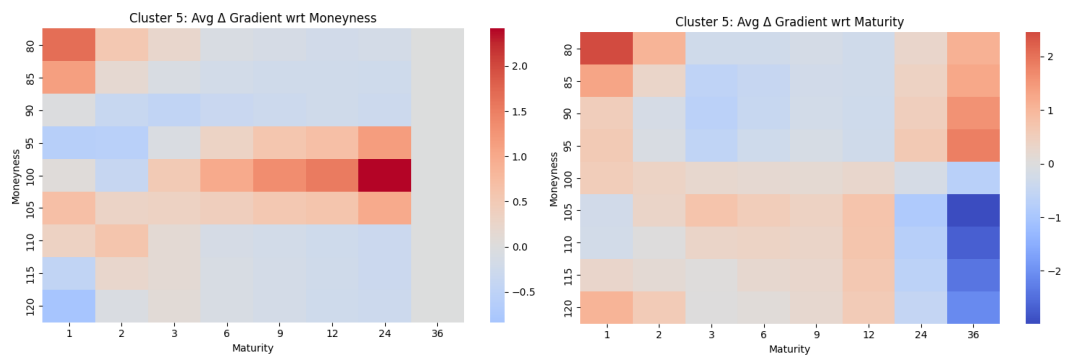
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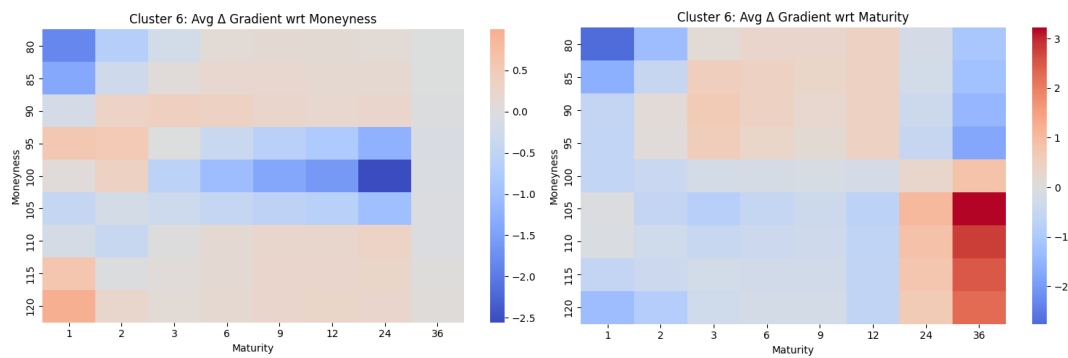
Cluster 7



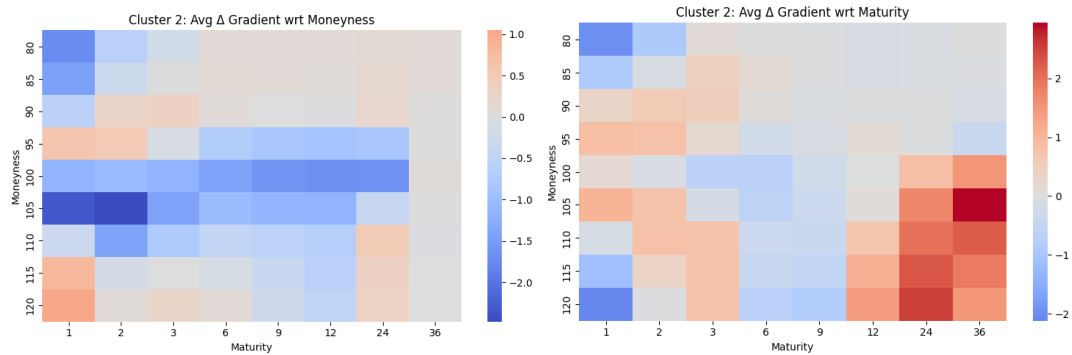
Cluster 8



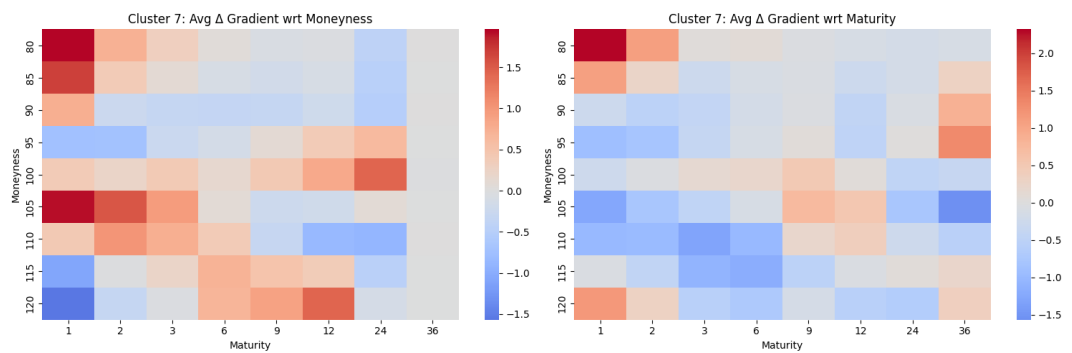
Cluster 9



Cluster 10



Cluster 11



Cluster 12

